

MID-SEMESTER EXAMINATION, SEPTEMBER 2022

Course Code: ECECC08/EIECC08

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Time: 1:30 Hours

Max. Marks: 15

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q.No.	Question	Marks	CO
1	Simplify the following Boolean function F with the don't-care conditions d and express in the SOP form $F(A, B, C, D) = \Sigma (5, 6, 7, 12, 14, 15)$ and $d(A, B, C, D) = \Sigma (3, 9, 11)$ (a) Express in the SOP form (b) Express in the POS form	1.5+ 1.5	CO1
2	(a) Implement the expression obtained in Q1 using NOR gates only. (b) If $X=1$ in the logic equation $[X+Z\{Y'+(Z'+XY')\}]\{X'+Z'(X+Y)\}=1$, then determine the value of Z.	1.5+ 1.5	CO3 CO1
3	Reduce the following Boolean expressions using boolean laws (a) $F(w, x, y, z) = \overline{(xy + z)} + z + xy + wz$ (b) $F(A, B, C, D) = AB(D + CD) + B(A + ACD)$	1.5+ 1.5	CO1
4	Perform subtraction on the given numbers (a) $100010.001 - 111.11$ using the 2's complement (b) $163.25 - 745$ using the 9's complement	1.5+ 1.5	CO1
5	(a) Draw a basic CMOS inverter circuit. (b) Find the Maxterms of the following expression $F(w, x, y, z) = (w+x')(w'+y+z')(x+y'+z')$	1.5+ 1.5	CO2 CO1

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23/9/22 (I)FN

Total no. of pages 1
B.Tech. III Semester (EE/ICE)

Roll no. _____

MID-SEMESTER EXAMINATION, SEPTEMBER 2022

Course Code: ICECC08/ EEECC08

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Time: 1:30 Hours

Max. Marks: 25

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q.No.	Question	Marks	CO
1	Simplify the following Boolean function F with the don't-care conditions d and express in the SOP form $F(A, B, C, D) = \Sigma (5, 6, 7, 12, 14, 15)$ and $d(A, B, C, D) = \Sigma (3, 9, 11)$ (a) Express in the SOP form (b) Express in the POS form	2.5+ 2.5	CO1
2	(a) Implement the expression obtained in Q1 using NOR gates only. (b) If $X=1$ in the logic equation $[X+Z\{Y'+(Z'+XY')\}]\{X'+Z'(X+Y)\}=1$, then determine the value of Z.	2.5+ 2.5	CO3 CO1
3	Reduce the following Boolean expressions using boolean laws (a) $F(w, x, y, z) = \overline{(xy + z)} + z + xy + wz$ (b) $F(A, B, C, D) = AB(D + CD) + B(A + ACD)$	2.5+ 2.5	CO1
4	Perform subtraction on the given numbers (a) $100010.001 - 111.11$ using the 2's complement (b) $163.25 - 745$ using the 9's complement	2.5+ 2.5	CO1
5	(a) Draw a basic CMOS inverter circuit. (b) Find the Maxterms of the following expression $F(w, x, y, z) = (w+x')(w'+y+z')(x+y'+z')$	2.5+ 2.5	CO2 CO1

Total No. of Pages

B.TECH 3rd Semesters

END SEMESTER EXAMINATION, DECEMBER 2021

Roll No. _____

(2)

Course Code: ICECC08/ECECC08/EIECC08/EEEECC08

Course Title: Digital Circuits & Systems

Time: 3 Hrs

Max Marks: 40

Note: Attempt all the five questions. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

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Q. No.	Question	Marks	CO
Q1	Attempt any 2 parts of the following	4	CO1
1a	Convert the following unsigned numbers: i. $(10110.11)_2 = ()_8$ ii. $(65477)_{10} = ()_{16}$ iii. Given $\sqrt{(224)}_r = (13)_r$. Calculate the value of the radix r.	4	CO1
1b	Perform the following operations using 2's complement method (a) $48 - 23$ (b) $23 - 48$	4	CO1
1c	What are the voltage and current interfacing equations for CMOS and TTL interfacing? Calculate the Fan-out for TTL logic family when $I_{OH} = 400 \mu A$, $I_{OL} = 16 \mu A$, $I_{IH} = 40 \mu A$, $I_{IL} = 1.6 \mu A$.		
Q2	Attempt any 2 parts of the following		
2a	Design a combinational circuit with three inputs X, Y and Z & three outputs A, B and C and explain its functioning?	4	CO2
2b	Describe Quine Mc Cluskey Tabulation method for solution of logic functions of four variables, write one advantage over K-map.	4	CO2
2c	Draw and explain the combinational network required to implement BCD to seven segment display.	4	CO2
Q3	Attempt any 2 parts of the following		
3a	Convert a D flip-flop to SR flip-flop.	4	CO3
3b	Draw circuit diagram for three bit up counter and explain its functioning.	4	CO3
3c	Implement a modulo 6 synchronous down-counter using T flip-flops.	4	CO3
Q4	Attempt any 2 parts of the following		
4a	What is hardware description (HDL) language, describe its role in digital design?	4	CO4
4b	Write VHDL code for an AND gate digital circuit.	4	CO4
4c	Describe role of entity and architecture of simple VHDL code	4	CO4
Q5	Attempt any 2 parts of the following		
5a	Write a short notes any two of the following: (i) FPGA (ii) PAL (iii) FGPA (iv) CPLD	4	CO5

5b	Explain SRAM and DRAM with the help of circuit diagrams. Also write down the differences between SRAM and DRAM.	4	COS
5c	Implement the following functions using PAL. $W = \Sigma(2, 12, 13)$ $X = \Sigma(7, 8, 9, 10, 11, 12, 13, 14, 15)$ $Y = \Sigma(0, 2, 3, 4, 5, 6, 7, 8, 10, 11, 15)$ $Z = \Sigma(1, 2, 8, 12, 13)$	4	COS

MID-SEMESTER EXAMINATION, SEPTEMBER 2023

Course Code: ECECC08/EIECC08

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Time: 1:30 Hours

Max. Marks: 15

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q.No.	Question	Marks	CO
1	A combinational circuit with three inputs, x, y, and z, and three outputs, A, B, and C. When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When The binary input is 4, 5, 6, or 7, the binary output is two less than the input. (a) Draw the truth table. (b) Implement by using NAND gates only	1+ 2	CO1
2	The following Boolean function $F(A, B, C, D) = \Sigma(1, 2, 5, 7, 8, 10, 11, 13, 15)$ (a) Draw the truth table (b) Implement it by using 4X1 MUX and external gates.	0.5+ 2	CO3
3	(a) Convert 110011 from gray code to binary. (b) Perform subtraction using 2's complement 11000.001 - 110.11	1+ 2	CO1
4	(a) Draw the circuit of CMOS two input NAND gate. (b) Represent the function in Canonical standard Sum of Product form. $F(w, x, y, z) = (w + x + y)(\bar{w} + xz) + \bar{y}z$	1.5+ 1.5	CO2 CO1
5	Design a BCD to Excess 3 decimal code convertor. (a) Make a truth table (b) Implement it by using a decoder of suitable size and minimal number of inputs to gates.	1+ 2	CO3

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Total no. of pages 1
B.Tech. III Semester (ECE/EIoT)

Roll no. _____

MID-SEMESTER EXAMINATION, SEPTEMBER 2023

Course Code: ECECC08/EIECC08

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Time: 1:30 Hours

Max. Marks: 15

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q.No.	Question	Marks	CO
1	A combinational circuit with three inputs, x, y, and z, and three outputs, A, B, and C. When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When The binary input is 4, 5, 6, or 7, the binary output is two less than the input. (a) Draw the truth table. (b) Implement by using NAND-gates only	1+ 2	CO1
2	The following Boolean function $F(A, B, C, D) = \Sigma(1, 2, 5, 7, 8, 10, 11, 13, 15)$ (a) Draw the truth table (b) Implement it by using 4X1 MUX and external gates.	0.5+ 2	CO3
3	(a) Convert 110011 from gray code to binary. (b) Perform subtraction using 2's complement 11000.001 - 110.11	1+ 2	CO1
4	(a) Draw the circuit of CMOS two input NAND gate. (b) Represent the function in Canonical standard Sum of Product form. $F(w, x, y, z) = (w + x + y)(\bar{w} + xz) + \bar{y}\bar{z}$	1.5+ 1.5	CO2 CO1
5	Design a BCD to Excess 3 decimal code convertor. (a) Make a truth table (b) Implement it by using a decoder of suitable size and minimal number of inputs to gates.	1+ 2	CO3

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(Shift-II)

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B.Tech. III Semester (EE/ICE)

Roll no. _____

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MID-SEMESTER EXAMINATION, SEPTEMBER 2023

Course Code: ICECC08/ EEECC08

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Time: 1:30 Hours

Max. Marks: 25

Note: Attempt all questions. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q.No.	Question	Marks	CO
1	A combinational circuit with three inputs, x, y, and z, and three outputs, A, B, and C. When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When The binary input is 4, 5, 6, or 7, the binary output is two less than the input. (a) Draw the truth table. (b) Implement by using NAND gates only	2+ 3	CO1
2	The following Boolean function $F(A, B, C, D) = \Sigma(1, 2, 5, 7, 8, 10, 11, 13, 15)$ (a) Draw the truth table (b) Implement it by using 4X1 MUX and external gates.	1+ 4	CO3
3	(a) Convert 110011 from gray code to binary. (b) Perform subtraction using 2's complement 11000.001 - 110.11	2+ 3	CO1
4	(a) Draw the circuit of CMOS two input NAND gate. (b) Represent the function in Canonical standard Sum of Product form. $F(w, x, y, z) = (w + x + y)(\bar{w} + xz) + \bar{y}\bar{z}$	2.5+ 2.5	CO2 CO1
5	Design a BCD to Excess 3 decimal code convertor. (a) Make a truth table (b) Implement it by using a decoder of suitable size and minimal number of inputs to gates.	2+ 3	CO3

END-SEMESTER EXAMINATION, NOVEMBER 2023

Course Code: ICECC08/ EEECC08

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Time: 3:00 Hours

Max. Marks: 50

Note: Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q.No.	Question	Marks	CO
1	Attempt any TWO parts (a) Design a combinational logic circuit that detects the presence of any of the six illegal code groups in the 2421 code by providing a logic-1 output. Implement it by using NOR gates only. (b) Explain the following parameters: (i) Figure of merit (ii) Fan-in (iii) Power supply requirements (iv) Noise Margin (c) Draw the circuit diagram for a CMOS circuit that realizes the Boolean expression $f(w, x, y, z) = [(w + x)\bar{y} + \bar{y}z]$	5 5 5	CO1 CO2 CO2
2	Attempt any TWO parts (a) Using the Quine-McClusky (Tabulation) method minimize the function given below having don't care terms $f(w, x, y, z) = \sum_m (1, 5, 6, 12, 13, 14) + d(0, 2, 4)$ (b) Design an multiplier circuit which multiplies two numbers of 2 bit each. Implement it by using a suitable size decoder and OR/NOR gates. (c) Implement the function using MUX of size 4 X 1, and external gates.	5 5 5	CO1 CO3 CO3

$$F(a, b, c, d) = a\bar{b} + acd + b\bar{c} + d$$

3

Attempt any TWO parts

(a) Design a register, incorporating four multiplexers and four positive edge triggered D flip-flop, having the behavior specified by table 1.

Select	lines	Register operation
S_1	S_2	
0	0	Hold
0	1	Synchronous preset
1	0	Complement contents
1	1	Parallel input

Table 1: Table for shift register

(b) Using a positive edge triggered J-K flip flop design a synchronous mod-5 counter whose counting sequence is

001 → 000 → 011 → 111 → 100 → 001

(c) Write short notes on Mealy and Moore sequential models.

4

Attempt any TWO parts

(a) Write VHDL code for the circuit that compares two numbers of 4 bit each.

(b) Write a VHDL code for a 4 bit UP/DOWN counter having clear and preset inputs.

(c) Write a VHDL code to implement 4X1 MUX using (i) behavioral and (ii) data flow modeling.

5

Attempt any TWO parts

(a) Design an adder that adds two numbers of two bit each and implement it by using PLA having both true and complemented outputs. Use the PLA of minimal number of product terms and draw the PLA table.

(b) Explain in brief ALU.

(c) Write a VHDL code for PWM Generator with Variable Duty Cycle.

END-SEMESTER EXAMINATION, NOVEMBER 2023

Course Code: ECECC08/EIECC08
Course Title: DIGITAL CIRCUITS AND SYSTEMS

Max. Marks: 40

Time: 3:00 Hours

Note: Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q.No.	Question	Marks	CO
1	Attempt any TWO parts (a) Design a combinational logic circuit that detects the presence of any of the six illegal code groups in the 2421 code by providing a logic-1 output. Implement it by using NOR gates only. (b) Explain the following parameters: (i) Figure of merit (ii) Fan-in (iii) Power supply requirements (iv) Noise Margin (c) Draw the circuit diagram for a CMOS circuit that realizes the Boolean expression $f(w, x, y, z) = [(w + x)\bar{y} + \bar{y}z]$	4 4 4	CO1 CO2 CO2
2	Attempt any TWO parts (a) Using the Quine-McClusky (Tabulation) method minimize the function given below having don't care terms $f(w, x, y, z) = \sum_m (1, 5, 6, 12, 13, 14) + d(0, 2, 4)$ (b) Design an multiplier circuit which multiplies two numbers of 2 bit each. Implement it by using a suitable size decoder and OR/NOR gates. (c) Implement the function using MUX of size 4 X 1, and external gates.	4 4 4	CO1 CO3 CO3

$$F(a, b, c, d) = a\bar{b} + acd + b\bar{c} + d$$

Attempt any TWO parts

(a) Design a register, incorporating four multiplexers and four positive edge triggered D flip-flop, having the behavior specified by table 1.

Select S_1	lines S_2	Register operation
0	0	Hold
0	1	Synchronous preset
1	0	Complement contents
1	1	Parallel input

Table 1: Table for shift register

(b) Using a positive edge triggered J-K flip flop design a synchronous mod-5 counter whose counting sequence is

001 $\rightarrow$ 000 $\rightarrow$ 011 $\rightarrow$ 111 $\rightarrow$ 100 $\rightarrow$ 001

(c) Write short notes on Mealy and Moore sequential models.

Attempt any TWO parts

(a) Write VHDL code for the circuit that compares two numbers of 4 bit each.

(b) Write a VHDL code for a 4 bit UP/DOWN counter having clear and preset inputs.

(c) Write a VHDL code to implement 4X1 MUX using (i) behavioral and (ii) data flow modeling.

Attempt any TWO parts

$$F(a, b, c, d) = a\bar{b} + acd + b\bar{c} + d$$

3

Attempt any TWO parts

(a) Design a register, incorporating four multiplexers and four positive edge triggered D flip-flop, having the behavior specified by table 1.

Select S_1	lines S_2	Register operation
0	0	Hold
0	1	Synchronous preset
1	0	Complement contents
1	1	Parallel input

Table 1: Table for shift register

(b) Using a positive edge triggered J-K flip flop design a synchronous mod-5 counter whose counting sequence is

001 $\rightarrow$ 000 $\rightarrow$ 011 $\rightarrow$ 111 $\rightarrow$ 100 $\rightarrow$ 001

(c) Write short notes on Mealy and Moore sequential models.

4

CO3

4

CO3

4

CO3

4

Attempt any TWO parts

(a) Write VHDL code for the circuit that compares two numbers of 4 bit each.

(b) Write a VHDL code for a 4 bit UP/DOWN counter having clear and preset inputs.

(c) Write a VHDL code to implement 4X1 MUX using (i) behavioral and (ii) data flow modeling.

4

CO4

4

CO4

4

CO4

5

Attempt any TWO parts

(a) Design an adder that adds two numbers of two bit each and implement it by using PLA having both true and complemented outputs. Use the PLA of minimal number of product terms and draw the PLA table.

(b) Explain in brief ALU.

(c) Write a VHDL code for PWM Generator with Variable Duty Cycle.

4

CO5

4

CO5

4

CO5

Total No. of Page(s): ...1....

Roll No.....

II- SEMESTER**B.E. (EC)**

MID SEMESTER EXAMINATION, February - 2020

ECC03: Digital Circuit and Systems

Time: 01:30 Hours

Max. Marks: 15

Note: Attempt all questions. Assume suitable missing data, if any.**Q1. (i)** Perform the subtraction $645_{10} - 319_{10}$ in excess-3 using 9's complement method. [1]**(ii)** Represent the function in Canonical standard SOP form

$$F(A, B, C, D) = (A + B + C)(\bar{A} + BD) + (\bar{C}\bar{D})$$

also find the minterms.

[2]

Q2. (i) Design a Full Adder using ROM.

[1]

(ii) Implement the following function

$$F(ABCD) = \sum m(0, 1, 3, 4, 7, 8, 9, 11, 14, 15)$$

using

[4]

(a) 4X1 MUX (Assume A, B are connected to the select line S_1 and S_0 respectively)**(b)** 2X1 MUX (Assume D is connected to the select line S_0)**Q3.** Minimize the expression using Quine McCluskey method.

[4]

$$Y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}CD$$

Q4. Design and implement excess-3 to BCD code converter using basic gates.

[3]

Date:

Roll No.

B.E IV SEMESTER ICE

MID-SEMESTER EXAMINATION 2020

ICC14 DIGITAL CIRCUITS AND SYSTEMS

Time: 1:30 Hrs

M.M.: 25

Attempt all questions. All questions carry equal marks. Assume missing data if any

Q[1] Verify using theorems and postulate

$$A \oplus B \oplus AB = A + B$$

Q[2] A switching network has four input $ABCD$. A and B represent the first and second bits of binary number N_1 . C and D represent the first and second bits of binary number N_2 . The output of the network is to be 1 only if the product $N_1 \times N_2$ is less than or equal to 2. Implement the network using 4×1 MUX and external gates.

Q[3] Design and implement a decimal code convertor that converts excess-3 code into BCD code using complementary output decoder of suitable size and AND/NAND gates.

Q[4] Realize the following functions using PLA having both true and complemented outputs. Use the PLA of minimal number of product terms and draw the PLA table.

$$\begin{aligned}f_1(x, y, z) &= \sum_m(1, 3, 4, 6) \\f_2(x, y, z) &= \sum_m(0, 2, 4, 5, 7) \\f_3(x, y, z) &= \prod_M(0, 2, 4)\end{aligned}$$

Q[5] Convert a negative edge triggered D flip-flop into negative edge triggered JK flip flop.